

**Lab sheet 09****Submission Due: 11.07.2024****Title:** PHP, MySQL and JavaScript - II**Aims:**

- Implement CRUD operations with PHP, MySQL, and JavaScript

**Tasks:**

- Implement CRUD operations in PHP
- Use fetch() to handle frontend requests

**Activities :**

1. Create a new table named “**expenses**” within “**expense\_tracker**” database

```
CREATE TABLE expenses (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    amount DECIMAL(10, 2) NOT NULL,  
    description TEXT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

2. Implement CRUD operation in PHP
  - I. add\_expense.php

```
<?php  
require_once '../database/Expense.php';  
require_once '../database/Database.php';  
  
$database = new Database("localhost", "root", "", "expense_tracker");  
$db = $database->getConnection();  
  
$expense = new Expense($db);  
  
$data = json_decode(file_get_contents("php://input"));  
  
$expense->setDescription($data->description);  
$expense->setAmount($data->amount);  
  
if ($expense->create()) {  
    echo json_encode(["message" => "Expense added."]);  
} else {  
    echo json_encode(["message" => "Unable to add expense."]);  
}  
?>
```

## II. view\_expenses.php

```
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';

$database = new Database("localhost", "root", "", "expense_tracker");
$db = $database->getConnection();

$expense = new Expense($db);

$expenses = $expense->getAllExpenses();
$num = count($expenses);

if ($num > 0) {
    $expenses_arr = array();
    $expenses_arr["records"] = $expenses;

    echo json_encode($expenses_arr);
} else {
    echo json_encode(["records" => []]);
}

?>
```

## III. edit\_expense.php

```
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';

$database = new Database("localhost", "root", "", "expense_tracker");
$db = $database->getConnection();

$expense = new Expense($db);

$data = json_decode(file_get_contents("php://input"));

$expense->setId($data->id);
$expense->setDescription($data->description);
$expense->setAmount($data->amount);

if ($expense->update()) {
    echo json_encode(["message" => "Expense updated."]);
} else {
    echo json_encode(["message" => "Unable to update expense."]);
}

?>
```

## IV. delete\_expense.php

```
<?php
require_once '../database/Expense.php';
require_once '../database/Database.php';

$databse = new Database("localhost", "root", "", "expense_tracker");
$db = $databse->getConnection();

$expense = new Expense($db);

$data = json_decode(file_get_contents("php://input"));

$expense->setId($data->id);

if ($expense->delete()) {
    echo json_encode(["message" => "Expense deleted."]);
} else {
    echo json_encode(["message" => "Unable to delete expense."]);
}
?>
```

## 3. Use fetch() to handle frontend requests

## I. loadExpenses()

```
async function loadExpenses() {
    try {
        const response = await fetch('php/view_expenses.php');
        const data = await response.json();
        const expenseList =
document.getElementById('expenseList');
        expenseList.innerHTML = '';

        if (Array.isArray(data.records) && data.records.length >
0) {
            data.records.forEach(expense => {
                const item = document.createElement('li');
                item.innerHTML = `${expense.created_at}:
${expense.description} - ${expense.amount}
                <button onclick="editExpense(${expense.id},
'${expense.description}', ${expense.amount},
'${expense.created_at}')">Edit</button>
                <button
onclick="deleteExpense(${expense.id})">Delete</button>`;
                expenseList.appendChild(item);
            });
        } else {
```

```
        const item = document.createElement('li');
        item.innerHTML = 'No expenses found.';
        expenseList.appendChild(item);
    }
} catch (error) {
    console.error('Error loading expenses:', error);
}
}
```

## II. addExpense()

```
async function addExpense() {
    const description =
document.getElementById('description').value;
    const amount = document.getElementById('amount').value;

    const response = await fetch('php/add_expense.php', {
        method: 'POST',
        body: JSON.stringify({ description, amount }),
        headers: { 'Content-Type': 'application/json' }
    });
    const result = await response.json();
    alert(result.message);
    loadExpenses();
}
```

## III. updateExpense()

```
async function updateExpense() {
    const id = document.getElementById('expenseId').value;
    const description =
document.getElementById('editDescription').value;
    const amount = document.getElementById('editAmount').value;

    const response = await fetch('php/edit_expense.php', {
        method: 'POST',
        body: JSON.stringify({ id, description, amount }),
        headers: { 'Content-Type': 'application/json' }
    });
    const result = await response.json();
    alert(result.message);
    document.getElementById('editForm').style.display = 'none';
    loadExpenses();
}
```

## IV. deleteExpense()

```
async function deleteExpense(id) {  
    const response = await fetch('php/delete_expense.php', {  
        method: 'POST',  
        body: JSON.stringify({ id }),  
        headers: { 'Content-Type': 'application/json' }  
    });  
    const result = await response.json();  
    alert(result.message);  
    loadExpenses();  
}
```

**Discussion:**

- *fetch()* function in JavaScript
- *mysqli\_fetch\_assoc()* function in PHP